Bonus Task 1: ECB vs CBC on same Image

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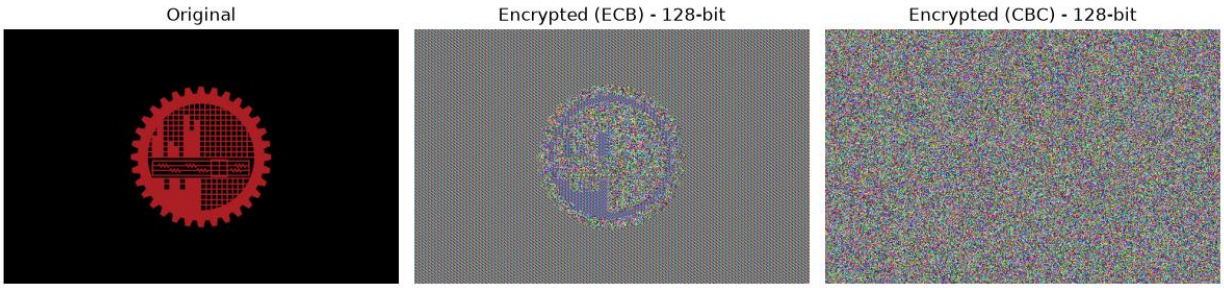
Objective

The same image was encrypted using two different AES modes of operation — ECB (Electronic Codebook) and CBC (Cipher Block Chaining) — using the same key. The raw pixel bytes of the image were treated as the plaintext. The goal is to visually compare the two outputs and explain the difference.

Method

A small image was loaded and converted to raw RGB pixel bytes. These bytes were encrypted independently with AES in ECB mode and in CBC mode (with a random IV), using the same key in both cases. The resulting ciphertext bytes were reinterpreted as pixel data and rendered back as images for visual comparison.

Output



Observation & Explanation

As seen in the output above, the ECB-encrypted image still reveals the outline and shape of the original image, while the CBC-encrypted image looks like random noise with no visible structure.

This happens because of how each mode processes blocks:

ECB Mode : Every 16-byte block of plaintext is encrypted independently using the same key, with no dependency on any other block. As a result, identical plaintext blocks always produce identical ciphertext blocks. Images typically contain large regions of flat or repeating color (background, uniform areas), which translate into many repeated 16-byte plaintext blocks. Since ECB maps identical inputs to identical

outputs, these repeated blocks stay visually consistent in the ciphertext, so the overall outline of the image remains recognizable even after encryption.

CBC Mode: Before encryption, each plaintext block is XORed with the previous ciphertext block (the first block is XORed with a random IV). This means every block's output depends on all the blocks before it. Even if two plaintext blocks are identical, their ciphertext blocks will differ, because each is chained to a different preceding context. This chaining effect removes the repeating-pattern leakage seen in ECB, so the CBC output has no visible structure and looks like random noise.

Conclusion

This demonstrates a well-known weakness of ECB mode — it does not hide data patterns, making it unsuitable for encrypting structured data such as images. CBC mode, through block chaining and the use of a random IV, avoids this weakness.